Teaching Philosophy

My goal as an educator is to help students understand the computational tools they use and develop the judgment to choose, question, and extend them. Strong foundations and intuitive understanding support this progression toward independent reasoning. Equally important is the ability to explain ideas, learn from others, and contribute to collaborative work. My educational experience has centered on mentoring undergraduate and master's students and engaging in research discussions with PhD students. These experiences inform my plans for classroom teaching and my approach to supporting students as they become independent researchers.

Foundations and understanding. A strong understanding of programming is essential to thoughtful use of AI tools. Data structures, algorithms, and computational complexity give students the means to evaluate generated code, compare possible solutions, and recognize limitations that successful execution alone may not reveal. In my future teaching, I will emphasize the reasoning behind implementation choices, including correctness, efficiency, and suitability for the problem. Asking students to explain these choices will connect the use of AI assistance with responsibility for understanding the resulting solution. Such foundations also prepare students to investigate research problems for which an apparently plausible answer requires further scrutiny.

I plan to introduce challenging concepts through concrete, motivating problems and a clear account of the central idea before presenting formal definitions or mathematical derivations. This sequence gives the formal treatment a purpose students can recognize. For example, a discussion of data structures can begin with the operations a problem requires, then examine how alternative representations affect implementation and computational cost. In machine learning, the problem a method addresses and the intuition behind its design can provide an entry point into its mathematical formulation. Returning to the motivating problem will help students connect abstract reasoning with practical decisions.

Supporting students. My plans for teaching include office hours where students can discuss difficulties openly and receive guidance responsive to their understanding. Listening carefully will be central to identifying whether a difficulty concerns a particular concept, an underlying assumption, or an unfamiliar way of approaching a problem. I also intend to be attentive to students navigating the transition to university and international students adjusting to a new educational and cultural environment. Recognizing differences in preparation and expectations about seeking help will inform how I support their participation and learning.

Teaching plans. My teaching interests include undergraduate courses in introductory programming, data structures, and software engineering, and graduate courses in machine learning and deep learning. Across these subjects, I intend to connect conceptual understanding with implementation and to adapt explanations to students' preparation. As students gain experience, the emphasis can expand from understanding how a solution works to evaluating assumptions, comparing approaches, and justifying technical decisions.

I am particularly interested in research-oriented courses that introduce the complete research process: understanding the literature, identifying a question, developing and implementing an approach, evaluating the evidence, and communicating the findings. Connecting these activities within a course will help students understand how research decisions depend on one another. It will also provide opportunities to consider how prior work or unexpected results can lead to a revised question or approach. The goal is to develop students' ability to form a reasoned position and identify the evidence needed to support it.

Research Mentorship

I adapt my mentoring to each student's knowledge, experience, and current research needs. Understanding how a student reasons about a problem helps me determine when to provide direct guidance and when to encourage greater ownership. The following experiences reflect different forms of support within that approach.

Entering research. My mentoring has included introducing the field, explaining the structure of research papers, and discussing the research process, alongside guided paper reading and pair programming. Reading together allows us to connect unfamiliar terminology and technical details with a paper's main argument, clarify misunderstandings, and establish a shared understanding of the work. Pair programming connects a method's description with its implementation and makes it possible to examine coding decisions together. These activities provide concrete starting points for engaging with research ideas.

Developing research questions. Through joint critical reading, we examine the problem a paper addresses, the assumptions underlying its approach, and the questions that remain unresolved. I have guided students in identifying open problems through these discussions, with an emphasis on understanding how a proposed direction relates to existing work and why it merits investigation. Research discussions with PhD students have also focused on higher-level ideas, problem framing, and theoretical insights. My contributions involve examining the reasoning behind an idea and helping sharpen the question under investigation. These exchanges extend the same emphasis on clear understanding to questions that require deeper conceptual analysis and more independent judgment.

Learning from results. Regular one-to-one meetings, typically weekly, are part of my mentoring practice. Students explain their progress and the evidence they have gathered, including positive findings and experiments that did not work as expected. Together, we interpret what the results reveal, which assumptions they support or challenge, and what to investigate next. I emphasize learning from unsuccessful experiments so that students can use the evidence to guide their next decision.

Independence and collaboration. As a future group leader, I intend to encourage collaboration among PhD students, creating opportunities to learn about developments in other areas and engage with different perspectives. Explaining an idea clearly, listening carefully, responding to constructive feedback, and contributing to a shared goal are important research skills. I want students to develop these abilities alongside their technical expertise and carry them into collaborations throughout their careers. My aim across teaching and mentorship is to support researchers who can reason independently while learning from and contributing to the work of others.